juliacon

MLThermoProperties.jl: Hybrid Models for Thermodynamic Property Prediction in Julia

Sebastian Schmitt[1], Hans Hasse[1], and Fabian Jirasek[1]

[1]Laboratory of Engineering Thermodynamics (LTD), RPTU Kaiserslautern

ABSTRACT

We present MLThermoProperties.jl, a Julia package that provides a variety of state-of-the-art thermodynamic models that combine modern machine learning methods with physical knowledge. These hybrid models obey hard physical constraints while being more accurate and applicable to a wider scope of substances than established models. MLThermoProperties.jl is built upon the Clapeyron.jl package, leveraging its rich thermodynamic solver ecosystem. The MLThermoProperties.jl models significantly improve molecular property prediction in various applications in science and engineering, e.g., chemical process engineering, and can be coupled with Julia's rich ecosystem for scientific modeling and simulation.

Keywords

Julia, Thermodynamics, Machine Learning, Chemical Engineering

1. Introduction

Knowledge of thermodynamic properties of fluids is crucial in many fields of engineering and science, e.g., for developing new chemical and biotechnological processes, optimizing heat pumps, or designing carbon capture and storage technologies. However, experimental data on thermodynamic properties are notoriously scarce due to the high cost and effort of measurements, making reliable prediction models indispensable. In recent years, machine learning (ML) has emerged as a particularly promising approach to thermodynamic modeling [2].

These models, despite being developed using methods from machine learning, are thermodynamically consistent, i.e., they obey hard physical constraints. This consistency is either achieved by an appropriate architecture [11, 6] or by exploiting existing thermodynamic models to form new hybrid models [13, 4, 5]. Based only on the SMILES code of a substance [15] – a textual representation of the molecular structure – the MLPROP models predict multiple thermodynamic properties, outperforming established models in both accuracy and scope [6]. Among the covered properties are phase equilibria of pure substances and mixtures [12], which are central to chemical process design, as well as transport properties such as diffusion coefficients [13] that govern molecular mass transfer. The models utilize different ML architectures, including the chemical language model ChemBERTa [1] and molecular graph neural networks.

In Julia, a rich ecosystem for thermodynamics and chemical engineering already exists. 'Clapeyron.jl' [14], a mature and comprehensive thermodynamic package, is a central part of this ecosystem. It combines a large number of thermodynamic models (including equations of state (EoS) and Gibbs excess energy (g^E) models) with advanced and efficient solvers. Leveraging Julia's excellent extensibility, several packages extend 'Clapeyron.jl', e.g., 'EntropyScaling.jl' [10] for modeling transport properties, or 'Langmuir.jl' [9] for modeling adsorption.

This paper introduces 'MLThermoProperties.jl', which provides Julia implementations of hybrid thermodynamic models, based on 'Clapeyron.jl' – enabling the prediction of thermodynamic properties for any substance in Julia. 'MLThermoProperties.jl' not only provides implementations of existing models, but also serves as a central anchor point for the development of new models. Due to the excellent extensibility of Julia packages, 'MLThermoProperties.jl' substantially broadens the capabilities for modeling chemical processes, based on state-of-the-art molecular property prediction. The individual models are presented and their integration into the existing thermodynamic ecosystem is explained. Additionally, illustrative applications of these models in chemical process simulations are showcased using packages from the wider scientific-modeling ecosystem in Julia, particularly from the SciML organization, e.g., 'ModelingToolkit.jl' and 'DifferentialEquations.jl'.

2. Package overview

MLThermoProperties.jl implements several hybrid models and makes them accessible to the thermodynamic solvers from Clapeyron.jl. This integration simplifies the implementation of new models and enables making use of the existing thermodynamic solvers already implemented in Clapeyron.jl.

2.1 Models

All currently implemented models were developed under the umbrella of MLPROP [8]. The following models are currently implemented and validated against their original publication:

HANNA Hard-constraint neural network for the excess Gibbs energy g^E of liquid mixtures, from which activity coefficients are obtained in a thermodynamically consistent manner [11]. Only the SMILES of the components and the temperature are required as input. Two versions are available: 'ogHANNA', the original model for binary mixtures [11], and 'HANNA' (alias 'multHANNA'), which is trained on VLE and LLE data and applicable to multicomponent mixtures [7].

UNIFAC 2.0 Machine-learning-enhanced versions of the group-contribution methods UNIFAC and mod. UNIFAC (Dortmund) for predicting activity coefficients [4, 3]. Missing group-interaction parameters are completed by matrix completion, which extends the applicability and accuracy of the methods. Both are implemented directly in ‘Clapeyron.jl‘.

GRAPPA Graph neural network for the vapor pressure of pure components, predicting the parameters of the Antoine equation and thereby also boiling points [5]. The SMILES of the component is the only input.

ESE Hybrid model for binary diffusion coefficients at infinite dilution that incorporates the Stokes-Einstein equation, ensuring a physically consistent temperature dependence [13]. It requires the SMILES of solute and solvent as well as a viscosity model for the solvent.

ChemBERTa Chemical language model that encodes a SMILES into a molecular embedding [1], which serves as molecular input representation in HANNA. It is provided by the registered package ‘ChemBERTa.jl‘ and can also be used independently.

2.2 Integration in Julia’s Thermodynamic Ecosystem

All models are implemented by extending a property function from Clapeyron.jl or EntropyScaling.jl and making use of existing thermodynamic solvers. This is exemplarily explained for the `HANNA` model and the `ESE` model in the following. For the Gibbs excess energy model `HANNA`, a custom function `Clapeyron.gibbs_excess_energy(model::HANNA,p,T,z)` is defined which calculates the Gibbs excess energy G^E through a neural network (see Ref. [6]). This is the only thermodynamic property that needs to be defined. Derived quantities like the activity coefficients γ_i and the excess enthalpy H^E can be calculated based on the HANNA model by Clapeyron’s property functions, i.e. `Clapeyron.activity_coefficient(model::HANNA,p,T,z)` and `Clapeyron.excess(model::HANNA,p,T,z,enthalpy)`, without further ado. This exploits the relation between those properties which can be calculated as derivations of G^E, i.e.

$$\ln \gamma_i = \frac{1}{RT} \left(\frac{\partial G^E}{\partial n_i} \right)_{T,p,n_{j \neq i}} \tag{1}$$

$$H^E = -RT^2 \left(\frac{\partial \left(G^E / RT \right)}{\partial T} \right)_{p,\boldsymbol{n}} . \tag{2}$$

The `ESE` model follows the same principle, but extends `EntropyScaling.jl` instead. The only method implemented evaluates the Stokes-Einstein equation corrected by a factor b_{ij}, which a neural network ensemble predicts from molecular descriptors of solute i and solvent j [13], i.e.

$$D_{ij}^{\infty} = b_{ij} \, \frac{k_{\mathrm{B}} T}{6\pi \, \eta_j(p,T) \, r_i} \tag{3}$$

with the Boltzmann constant k_{B} and the hydrodynamic radius r_i of the solute. Since b_{ij} depends on the components only, it is evaluated once when the model is constructed and the temperature dependence is dictated entirely by the Stokes-Einstein equation, which enforces physical consistency. The solvent viscosity η_j is obtained by calling `EntropyScaling.viscosity(model,p,T)` on a viscosity model supplied by the user, so that any model of `EntropyScaling.jl` can be combined with `ESE`, e.g., a substance-specific reference correlation (`RefpropRES`), a group-contribution entropy scaling model (`GCES`), or a constant experimental viscosity (`ConstantModel`). Based on this single method, `EntropyScaling.inf_diffusion_coefficient` is readily available for `ESE`, returning the diffusion coefficients of all component pairs or of an individual solute-solvent pair.

3. Example Application

3.1 Calculating Mixture Properties

Code 1: Example Code Block.

```julia
using Plots

x = -3.0:0.01:3.0
y = rand(length(x))
plot(x, y)
```

3.2 Chemical Process Simulation

4. Conclusions

5. Acknowledgements

6. References

[1] W. Ahmad, E. Simon, S. Chithrananda, G. Grand, and B. Ramsundar. ChemBERTa-2: Towards Chemical Foundation Models, 2022. doi:10.48550/arXiv.2209.01712.

[2] H. Hasse, S. Schmitt, and F. Jirasek. Artificial intelligence in thermodynamics: hybrid modeling of thermophysical properties of fluids. *Current Opinion in Chemical Engineering*, 51:101236, 2026. doi:10.1016/j.coche.2026.101236.

[3] N. Hayer, H. Hasse, and F. Jirasek. Modified UNIFAC 2.0-A Group-Contribution Method Completed with Machine Learning. *Industrial & Engineering Chemistry Research*, 64(20):10304–10313, 2025. doi:10.1021/acs.iecr.5c00077.

[4] N. Hayer, T. Wendel, S. Mandt, H. Hasse, and F. Jirasek. Advancing thermodynamic group-contribution methods by machine learning: UNIFAC 2.0. *Chemical Engineering Journal*, 504:158667, 2025. doi:10.1016/j.cej.2024.158667.

[5] M. Hoffmann, H. Hasse, and F. Jirasek. GRAPPA—A hybrid graph neural network for predicting pure component vapor pressures. *Chemical Engineering Journal Advances*, 22:100750, 2025. doi:10.1016/j.ceja.2025.100750.

[6] M. Hoffmann, T. Specht, Q. Göttl, J. Burger, S. Mandt, H. Hasse, and F. Jirasek. A machine-learned expression for the excess Gibbs energy, 2025. doi:10.48550/arXiv.2509.06484.

[7] M. Hoffmann, T. Specht, Q. Göttl, J. Burger, S. Mandt, H. Hasse, and F. Jirasek. Thermodynamically consistent machine learning model for excess Gibbs energy. *Nature Communications*, 17(1):3485, 2026. doi:10.1038/s41467-026-71430-y.

[8] M. Hoffmann, T. Specht, N. Hayer, H. Hasse, and F. Jirasek. MLPROP - An Interactive Web Interface for Thermophysical Property Prediction with Machine Learning. *Chemie Ingenieur Technik*, 97(11-12):1052–1056, 2025. doi:10.1002/cite.70004.

[9] V. V. Santana, A. Riedemann, T. Christiansson, and P. J. Walker. Langmuir.jl.

[10] S. Schmitt, A. Riedemann, E. B. D. M. Alves, and LéoSchmider. se-schmitt/EntropyScaling.jl: v0.3.5, 2025. doi:10.5281/zenodo.17097480.

[11] T. Specht, M. Nagda, S. Fellenz, S. Mandt, H. Hasse, and F. Jirasek. HANNA: hard-constraint neural network for consistent

activity coefficient prediction. *Chemical Science*, 15:19777–19786, 2024. doi:10.1039/D4SC05115G.

[12] T. Specht, M. Nagda, S. Fellenz, S. Mandt, H. Hasse, and F. Jirasek. HANNA: Hard-constraint Neural Network for Consistent Activity Coefficient Prediction, 2024. doi:10.48550/arXiv.2407.18011.

[13] J. Wagner, Z. Romero, K. Münnemann, S. Schmitt, T. Specht, H. Hasse, and F. Jirasek. Hybrid Machine Learning for Enhanced Prediction of Diffusion Coefficients in Liquids, 2026. doi:10.48550/arXiv.2603.02761.

[14] P. J. Walker, H.-W. Yew, and A. Riedemann. Clapeyron.jl: An Extensible, Open-Source Fluid Thermodynamics Toolkit. *Industrial & Engineering Chemistry Research*, 61(20):7130–7153, 2022. doi:10.1021/acs.iecr.2c00326.

[15] D. Weininger. SMILES, a chemical language and information system. 1. Introduction to methodology and encoding rules. *Journal of Chemical Information and Computer Sciences*, 28(1):31–36, 1988. doi:10.1021/ci00057a005.